Translating TEI XML Documents Without Breaking Them

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1 Very brief summary

We want to build a translation tool for TEI XML files, the format used by historians and literature scholars to digitize old texts. Normal translation tools don't understand what the tags in these files mean, so they end up translating things that should stay untouched like XML tags, quotes in foreign languages and TEI-specific attributes. We want to build a tool that actually understands these tags and translates documents correctly.

2 Some details

Task type: This is a machine translation task, with special caveats: TEI documents contain meaningful tags around parts of the text that tell us how each part should be handled. Some parts should be translated normally, some should be left exactly as they are (like a quote deliberately kept in the original language), and some might need the alphabet changed instead of being translated (like a name written in Cyrillic that needs to become Latin script).

Why this is interesting: This has not been done before. Tools exist to translate other formats (Markdown, JSON, etc.). The TEI ecosystem already has poweful tools that display TEI nicely (see TEI Publisher), or convert TEI into PDF/HTML (see TEI Consortium's TEIGarage). But there is no tool that can translate TEI documents while respecting the meaning of the tags. We confirmed this ourselves: running a TEI document through DeepL and Google Translate wrongly translated a quote meant to stay untranslated, corrupted a person's name and changed tags.

Dataset: We plan to use real TEI files from the Perseus Digital Library, openly available on GitHub, which contains thousands of freely licensed Greek, Latin and English texts, several paired with scholar-made translations we can use as reference. There are many other digital libraries which will also be considered.

Our approach: A pipeline that takes a TEI file as input, applies rules for how each tag should be handled, uses alignment techniques for moving text when necessary, and produces a translated TEI file. Verifications (including TEI syntax) are done to the translation against the original to ensure validity.

Experiment: We will evaluate against both a naive baseline (like DeepL) and scholarly translations, measuring structural validity, correct tag handling, and translation quality.

Challenges: Defining correct handling rules for every tag type present in our chosen documents, since TEI's tag vocabulary is large and not all cases are obvious; ensuring the translated output remains valid, well-formed TEI; and correctly deciding, for names and places, whether transliteration or standard translation is appropriate. Word alignment is another NLP challenge: when a sentence is translated, word order changes between languages, so figuring out where a XML element, untouched or transliterated piece belongs is not a simple lookup, it requires alignment techniques used in machine translation research.

Open questions: What features will we implement in this tool? We have many ideas for the tool. For instance, TEI supports linking two or more documents, the tool could link passages from the original to the translation, improving multi-document viewing. Which features will we implement, and which techniques will we end up using is an open question.